T028 bond learning report

Project: T020 reviewed assembly
Analysis: T020 reviewed assembly · frozen revision 7
Run completed: 2026-10-16T04:38:46.835697+00:00
Report generated: 2026-10-16T04:39:00.584270+00:00
Display units: mm-N-MPa

Report ce26caa5-a09c-4598-9664-380f12403082
Run c9a4340a-37e7-4cfa-a571-8cd8f2deb986
Input SHA-256 1ae79079c932ada4c5f7d292dec699c3f3061c242e6a61cf29fccf84a6836326

Scope, warnings and assumptions

> Gravity unavailable: density is missing.

> Yield factor of safety unavailable: yield strength is missing.

> Connectivity and rigid-motion support checks are necessary only, not proof of stiffness rank or stability; no weak springs are added.

stress_singularity · all bodies · unassessed: Peaks near supports, reentrant corners or idealized load/bond edges may increase with refinement. No automatic singularity detection is claimed. Compare fixed physical locations away from these regions; a lower smoothed peak is not evidence of convergence or a design allowable.

yielding · 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1 · unassessed: Compare raw integration-point von Mises with the frozen supplied ductile yield. Yield exceedance does not model plastic deformation; missing yield cannot establish a factor. No fatigue, brittle failure, buckling or code check is implied.

small_deformation · 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1 · not triggered: Maximum native displacement / largest body extent is a 1% screening heuristic. Exceedance requires review of small-deformation linear elasticity. Passing this heuristic does not bound local strain/rotation or certify validity; display exaggeration is irrelevant.

yielding · 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2 · unassessed: Compare raw integration-point von Mises with the frozen supplied ductile yield. Yield exceedance does not model plastic deformation; missing yield cannot establish a factor. No fatigue, brittle failure, buckling or code check is implied.

small_deformation · 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2 · not triggered: Maximum native displacement / largest body extent is a 1% screening heuristic. Exceedance requires review of small-deformation linear elasticity. Passing this heuristic does not bound local strain/rotation or certify validity; display exaggeration is irrelevant.

Linear static, small-deformation and supplied material assumptions apply. No automatic design-safety conclusion is generated.

Yield assessment does not assess fatigue, brittle failure, buckling or plasticity. Stress singularities require engineering review.

Passing equilibrium does not establish convergence, stability or physical validity. Target-laptop validation and remaining external references are pending; bounded M3 approval does not certify this design.

Figures are light report derivatives of retained verified data. Hidden bodies, clipping, smoothing and deformation do not alter all-body native extrema or warnings.

Selected figures and setup

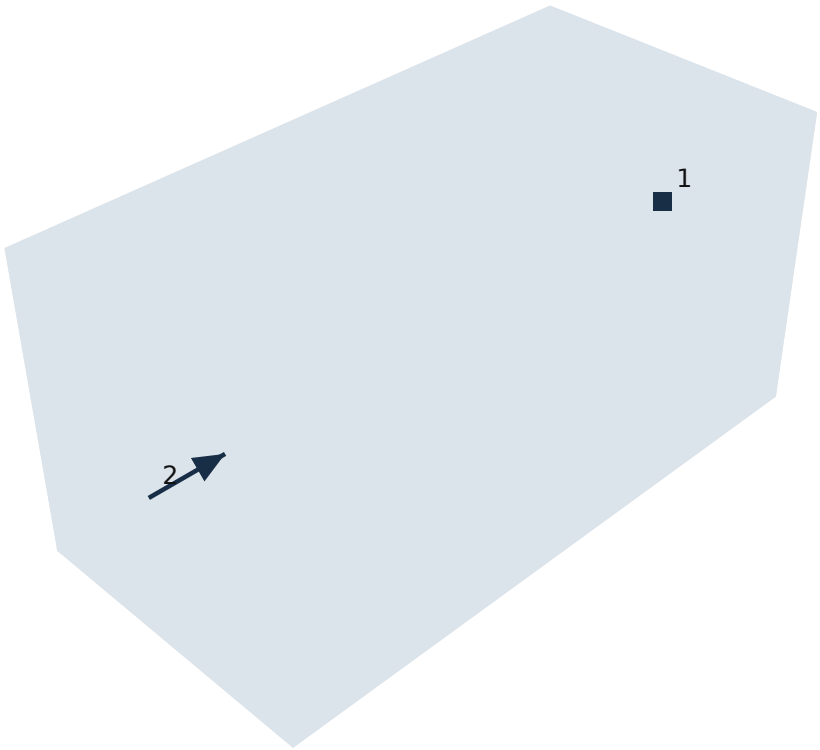
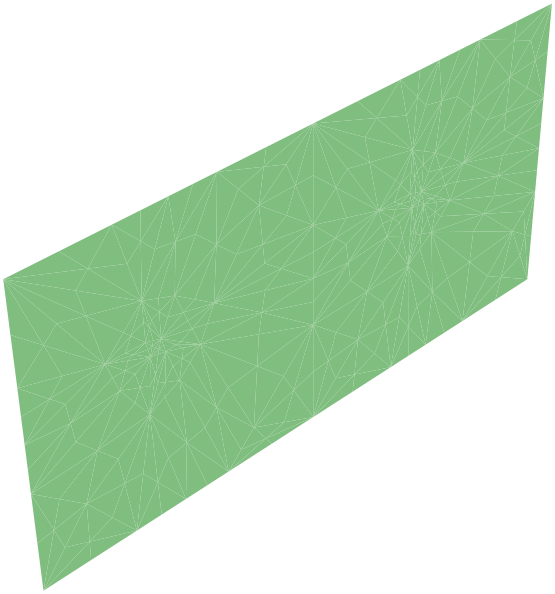


Figure 1. Frozen loads and supports on undeformed geometry. Arrows indicate global direction only; squares indicate fixed supports. Marker numbers resolve to the setup table; arrow lengths do not encode magnitude.



Stress σxx (MPa): -0.001 to -0.001

Constant field; uniform colour.

Figure 2. Stress σxx; global Cartesian; element-interpolated volume section {'point_m': [0.0, 0.5, 0.0], 'normal': [0.0, 1.0, 0.0]}; linear_C3D10_extrapolation_from_native_DAT_IP; element-owned tensors, then invariants; default unaveraged, optional body/material-partitioned smoothing. Deformation ×0; visible bodies 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1, 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2. Flat triangle colour samples use the vertex-scalar mean; no numerical extrema are read from pixels. Camera and SI limits are retained in the report model.

Geometry and source units

Fact	Frozen value
Geometry revision	0f18e552-5805-4bd0-9353-431601965e09
Source length unit	m
Bounds (mm)	[0, 0, 0, 2000, 1000, 1000]
Dimensions (mm)	[2000, 1000, 1000]
Bodies / faces	2 / 12
Geometry SHA-256	5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996

Material properties

Body / material	E (MPa) / Poisson ratio	Density (tonne/mm³)	Yield (MPa)
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1 / Synthetic axial reference	68300 / 0	Unavailable	Unavailable
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2 / Synthetic axial body two	68300 / 0	2.7e-09	Unavailable

Material provenance

Body	Detail	Source / assumption
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1	Grade / condition	Synthetic / Test only
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1	Temperature / assumptions	20 C reference / Isotropic analytical reference, E=68.3 GPa, nu=0. No certified allowable.
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1	Library / original units	None / None / mm-N-MPa
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1	Independent synthetic axial fixture	No URL supplied; accessed not supplied
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2	Grade / condition	6061 / T6/T651 rod and bar; typical properties
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2	Temperature / assumptions	Room-temperature example; density reference 20 °C / Independent analytical isotropic fixture: E=68.3 GPa, nu=0; no certified allowable.
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2	Library / original units	None / None / mm-N-MPa
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2	Independent axial continuum reference	No URL supplied; accessed not supplied

Loads and supports

Assignment	Immutable targets	Physical value / direction
Fixed root / fixed	5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1:f1	All translations fixed
End pressure / pressure	5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2:f2	Pressure MPa: 0.001

Setup figure markers

Marker	Assignment / kind	Position (mm) / global direction
1	0a8074d5-09fb-478c-98d3-573d07517ec5 / fixed	[0, 500, 500] / [0, 0, 0]
2	2e3de3b2-5754-462f-97e1-818b29d7afe5 / pressure	[2000, 500, 500] / [-1, 0, 0]

Bond review and gaps

Connection	Master / slave	Approved gap / review
Reviewed zero gap	5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1:f2 / 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2:f1	Measured 0 m; tolerance 0 m; gap approved False; assumptions confirmed True; policy planar-unit-complete-b08-v1; review SHA-256 a512632680af40de32cb18be0548060e48fde553e929e08a3f38bc6657eaea81c

Mesh and solver

Fact	Frozen value
Element / node count	C3D10 / 200 elements / 462 nodes
Quality	{"schema_version": 1, "accepted": true, "minimum_jacobian_m3": 0.032958984374994345, "minimum_mean_ratio": 0.6694380324434518, "bad_elements": [], "warning_elements": [], "all_affine": true, "policy": {"jacobian_floor_m3": 1e-15, "mean_ratio_floor": 0.05, "mean_ratio_warning": 0.2, "subdivision_depth": 3, "method": "cubic tetrahedral Bernstein
	certificate with outward rounding margin"}}

Fact	Frozen value
Mesh controls	{"schema_version": 1, "geometry_revision_id": "0f18e552-5805-4bd0-9353-431601965e09", "global_size_m": 0.6, "source_global_size": 600.0, "source_units": "mm-N-MPa", "local_sizes": [], "curvature_elements": 0, "recovery_enabled": false}
Solver	CalculiX 2.23
Solver binary SHA-256	d55622f40e0435e738a86cbd364d525fff31524212ecca17d71d4c15ea2935f5
Original solve CPU/RAM/thread inventory	Not recorded in this legacy local run; resource policy and measured usage are retained in run artifacts.
Native build SHA-256	917a915f2aa78656131eeba825c13dece6cf81834d786a317d7dc9e6ef6db74d
Numerical method	Linear static / small deformation / global Cartesian / C3D10; no stabilization

Raw native extrema — all bodies

Field / location	Minimum	Maximum / identity
Displacement magnitude (mm); node	0; node; node 1; element None; IP None; body None; material None; position [0, 0, 0] mm	2.928258e-05; node; node 13; element None; IP None; body None; material None; position [2000, 0, 0] mm
Von Mises stress (MPa); integration_point	0.001; integration_point; node None; element 343; IP 4; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 391.2994, 951.4079] mm	0.001; integration_point; node None; element 355; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 380.8701, 38.86779] mm
Displacement ux (mm); node	-2.928258e-05; node; node 13; element None; IP None; body None; material None; position [2000, 0, 0] mm	0; node; node 1; element None; IP None; body None; material None; position [0, 0, 0] mm
Displacement uy (mm); node	0; node; node 1; element None; IP None; body None; material None; position [0, 0, 0] mm	7.414123e-19; node; node 75; element None; IP None; body None; material None; position [2000, 250, 1000] mm
Displacement uz (mm); node	-3.258162e-19; node; node 15; element None; IP None; body None; material None; position [2000, 1000, 0] mm	8.782731e-22; node; node 135; element None; IP None; body None; material None; position [140.625, 0, 140.625] mm

Field / location	Minimum	Maximum / identity
Stress σxx (MPa); integration_point	-0.001; integration_point; node None; element 169; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1; material M1; position [284.1728, 158.3939, 284.1728] mm	-0.001; integration_point; node None; element 169; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1; material M1; position [284.1728, 158.3939, 284.1728] mm
Stress σyy (MPa); integration_point	-7.649496e-18; integration_point; node None; element 310; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1942.462, 460.0269, 699.8698] mm	5.375619e-18; integration_point; node None; element 355; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 380.8701, 38.86779] mm
Stress σzz (MPa); integration_point	-6.188083e-18; integration_point; node None; element 343; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 167.6926, 951.4079] mm	5.06538e-18; integration_point; node None; element 344; IP 4; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 948.1763, 170.2192] mm
Stress σxy (MPa); integration_point	-1.408342e-17; integration_point; node None; element 343; IP 4; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 391.2994, 951.4079] mm	1.045946e-17; integration_point; node None; element 310; IP 2; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1942.462, 524.1047, 857.1171] mm
Stress σyz (MPa); integration_point	-3.554738e-18; integration_point; node None; element 359; IP 4; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1963.247, 738.4629, 844.3137] mm	4.84884e-18; integration_point; node None; element 343; IP 4; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 391.2994, 951.4079] mm
Stress σzx (MPa); integration_point	-6.333143e-18; integration_point; node None; element 294; IP 3; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1893.87, 534.7047, 339.055] mm	9.417029e-18; integration_point; node None; element 343; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 167.6926, 951.4079] mm
Principal stress σ1 (MPa); integration_point	-2.018635e-18; integration_point; node None; element 302; IP 3; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1905.709, 576.2516, 905.7092] mm	6.44745e-18; integration_point; node None; element 355; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 380.8701, 38.86779] mm

Field / location	Minimum	Maximum / identity
Principal stress σ2 (MPa); integration_point	-9.010354e-18; integration_point; node None; element 343; IP 4; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 391.2994, 951.4079] mm	1.293288e-18; integration_point; node None; element 365; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1807.561, 926.4949, 807.5612] mm
Principal stress σ3 (MPa); integration_point	-0.001; integration_point; node None; element 169; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1; material M1; position [284.1728, 158.3939, 284.1728] mm	-0.001; integration_point; node None; element 169; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1; material M1; position [284.1728, 158.3939, 284.1728] mm

Yield assessment — frozen supplied strength / raw integration-point von Mises

Body / material	Yield (MPa)	Factor / critical location
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1 / Synthetic axial reference	Unavailable	Unavailable — missing yield; integration_point; node None; element 212; IP 4; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1; material M1; position [914.6554, 49.29717, 533.0508] mm
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2 / Synthetic axial body two	Unavailable	Unavailable — missing yield; integration_point; node None; element 355; IP 1; body 5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2; material M2; position [1951.408, 380.8701, 38.86779] mm

External reactions about global origin

Body / constrained nodes / DOFs	Force (N)	Moment (N·mm)
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b1 / 37 / 111	[1000, -8.168552e-13, 2.127771e-13]	[7.485757e-10, 500000, -500000]
5d8c2622a7778c7cd6373d8482491a964fb6751592aac73ea691b6b07fd33996:b2 / 0 / 0	[0, 0, 0]	[0, 0, 0]

Force and moment equilibrium

Balance	Residual	Reference	Criterion / limit	Outcome
force (N)	[9.999998e-06, -8.192639e-13, 2.421626e-13]; norm 9.999998e-06	1000	relative; relative 9.999998e-09; allowed 5	Passed
moment (N·mm)	[7.859354e-10, 0.003088851, -0.006630524]; norm 0.007314701	2000000	relative; relative 3.657351e-09; allowed 10000	Passed

Equilibrium method and actions

Fact	Value
Method	Frozen physical load magnitudes plus nonzero prescribed-component actuator reactions; moments about global origin; external constrained span only; internal ties/MPC/free-node RF excluded
Reaction method	DAT RF minus written equivalent CLOAD/pressure/consistent gravity, constrained global components only; unique nodes/DOFs per body; internal ties excluded
Moment origin (m)	[0, 0, 0]
Applied force (N)	[-1000, -2.408648e-15, 2.938551e-14]
Applied moment (N·mm)	[3.73597e-11, -500000, 500000]
Stress / extrema method	linear_C3D10_extrapolation_from_native_DAT_IP; element-owned tensors, then invariants; default unaveraged, optional body/material-partitioned smoothing; Displacement: all native DAT nodes; von Mises/yield factor: all raw DAT integration points, including interior

User-written conclusion

Attributed to the report author; this text is not a computed engineering finding.

No user conclusion supplied.

Reproducibility IDs

schema_version: 1

job_id: c9a4340a-37e7-4cfa-a571-8cd8f2deb986

analysis_id: 1546e05e-4194-431f-8e65-1b69625fb145

analysis_revision: 7

geometry_revision_id: 0f18e552-5805-4bd0-9353-431601965e09

input_sha256: 1ae79079c932ada4c5f7d292dec699c3f3061c242e6a61cf29fccf84a6836326

results_sha256: ba94751c1e4d4eb4a0ef439d93f2d7ee9e5b5e43a1bc55ca7f5935d29862ad8a

mesh_sha256: dfa27971b1a0c36d365399d4dd00e6b4d7898a21748cbe312ac6a96d96d2a162

deck_sha256: 0a02defedb8f7c5ee282107da20692214c8f4ed91ff130eb2ba474c8ce131637

dat_sha256: 00ab27e4a6987d88d6dbb9c4413468550eb758a0d8340e428d52e9cbb90cbd0a

services/worker/worker/postprocess/view.py: 7ea99bb6c7c9daf7277201e6779148eba316d236c0699a9629348e8900124a8c

services/worker/worker/postprocess/inspection.py: 2671965a0e9881217553f96e8389289d24f389b7b002569b370fe92a01486f08

services/worker/worker/postprocess/comparison.py: 8970cbe6b7e9686a77176bc4d8ae432217276bf4f2f57577c55c837941d8a3aa

services/api/app/domain/comparisons.py: 7031bcf9a45c54cef209210382718a5057449d41e29fded6bab882316e389bc0

services/api/app/domain/refinement.py: b63ca772fc862a035dda9f2c3f63d06326fa3cbda50f093e5efae4b4c258a996

services/worker/worker/postprocess/fields.py: 3201afd7ad773d0865f77b992e941c5264484d87490964b84866ff87e718c2cf

services/worker/worker/postprocess/tensors.py: 9dc5efb68ebfafc05d5f964019548b801fdcad8706032554334aa0c537767584

services/worker/worker/meshing.py: 1d71f8f57bd020f96804f284f4be1d2a0fd51bf89d0bc529f48cd1aad2a52e9e

services/worker/worker/actions.py: 1a8edff017a1397b45fa667d9056a2b5e9ebf250248812928855f64dec460e9b

services/worker/worker/directions.py: 8de17d5705b81b651a5077ddf3b80e3b29a13b9f453af1c79fc2e82db06e4592

services/worker/worker/calculix/results.py: da43f91763e9880a6e8728906202f1394d7b89a31af61e6a6f163c1190c29609

services/api/app/jobs/service.py: 638d211f21fa6e9d8b4dcce0e5a6fa5b5cace3c1ce1e2eee4e3131266965ae1e

services/api/app/domain/results.py: c4c0b9b18b090bea8352ea59a2a09776d3b20783598f81c0db6b6ce18706b3ad

version: report-v1

source_commit: d1e4a49b9aa347b4aa8e185c622ea44240b65022

source_dirty: true

generator_python: 3.12.14

generator_platform: Linux-6.18.44-x86_64-with-glibc2.41

generator_cpu_count: 3

font_dejavu_regular_sha256: 57f73e11f51999432bf7ab22ce55b6f945d5eca1bf824404cfa9ec2e3718c84e

chromium_binary_sha256: 4fa61f0ba0b4f89b7bec242c32ca16d3bccf69dbfef0651924a9ba8fede5f9d6

node_binary_sha256: bc17c508ffeed0ec622934f9b7fa72f8e78da65350e63c3eceb56fa688aa5e12

services/api/app/domain/reports.py: 45a4b8349f9ae22f3bb63d50f7ae8acce9e8129136a6969f2cbdfd8274789ca0

services/api/app/domain/report_service.py: 1e417c01670abd940a3d73f22e6555d96095dbfa88db782c6efc6380b42cd776

services/worker/worker/reports/model.py: adcb4b46695ab4ddad618402cd8d6bb613332fa0a2c3ddd97d6227d0caf7dd44

services/worker/worker/reports/figures.py: bedc9575b4b72309897e2eba8fc0664ac04e84b7de74615e1d595e29eca19047

services/worker/worker/reports/html.py: a266e66c2204f726528b7bc307935b2ef7096153fbbc58146ee5d7a7c1316bb9

services/worker/worker/reports/content.py: 5914c8af6cddf62125566746a8d881fcdb5a2ef8d40a84bc77095e248e929b3e

services/worker/worker/reports/render.py: 8b4f6ae521693827346fe2ab67de5e680cc583cdcdbf4d2780c6f70db0dd98de

scripts/reports/pdf.mjs: 125def705cc32a64a6ace989b65a573dac6af6de54fd02c1ccd11064ea65c123

package-lock.json: feb1470637caf3d65d9af3d08e89ca777ca097f49b055845452b610917c45fba

uv.lock: 36f59cf0e56b6442588403ba9ac6b47e732a46ce4a9d8cd7f3e5edcea741e261

Download the immutable report-model JSON for complete original inputs, view definitions, camera, quality and tool policy. Its SHA-256 and this PDF’s SHA-256 are in the report record.